RAJALAKSHMI ENGINEERING COLLEGE

Suriya Prakash D 251001277-2025-IT S2

Started on	Wednesday, 7 October 2026, 9:26 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:26 AM
Time taken	23 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x) {
    // No element <= x
    if (low > high)
        return -1;

    // If largest element is <= x
    if (arr[high] <= x)
        return arr[high];

    // If smallest element is greater than x
    if (arr[low] > x)
        return -1;

    int mid = low + (high - low) / 2;

    if (arr[mid] == x)
        return arr[mid];

    if (arr[mid] < x) {
        // arr[mid] can be the floor, but search for a larger one
        int result = findFloor(arr, mid + 1, high, x);

        if (result == -1)
            return arr[mid];

        return result;
    }

    // arr[mid] > x
    return findFloor(arr, low, mid - 1, x);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int result = findFloor(arr, 0, n - 1, x);

    if (result == -1)
        printf("No\n");
    else
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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